

Ex. No : 22

Date : Basics of Pandas

Aim

To understand the basics of Pandas and perform various data handling operations using Series and DataFrames.

Theory

Pandas is a Python library used for data manipulation and analysis. It provides powerful data structures such as Series and DataFrame, which help in handling and analyzing structured data efficiently.

22.1 Program to Create a Pandas Series

Program

```
import pandas as pd

s = pd.Series([10, 20, 30, 40, 50])

print(s)
```

Sample Output

```
0    10
1    20
2    30
3    40
4    50
dtype: int64
```

22.2 Program to Create a Pandas DataFrame

Program

```
import pandas as pd

student = {
    'Name': ['Amit', 'Riya', 'Rahul'],
    'Marks': [85, 90, 78]
}

_df = pd.DataFrame(student)

print(_df)
```

Sample Output

	Name	Marks
0	Amit	85
1	Riya	90
2	Rahul	78

22.3 Program to Read Data from a CSV File Using Pandas

Program

```
import pandas as pd

_df = pd.read_csv('data.csv')

print(_df)
```

Sample Output

	Name	Age
0	Amit	20
1	Riya	21

22.4 Program to Display First and Last Rows of a DataFrame

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', 'Rahul', 'Neha'],
    'Marks': [85, 90, 78, 88]
}

_df = pd.DataFrame(data)

print("First Rows:")
print(_df.head())

print("Last Rows:")
print(_df.tail())
```

Sample Output

```
First Rows:
   Name  Marks
0  Amit     85
1  Riya     90
2  Rahul    78
3  Neha     88
Last Rows:
   Name  Marks
0  Amit     85
1  Riya     90
2  Rahul    78
3  Neha     88
```

22.5 Program to Select Specific Columns from a DataFrame

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', 'Rahul'],
```

```
'Age': [20, 21, 22],
'Marks': [85, 90, 78]
}

_df = pd.DataFrame(data)

print(_df[['Name', 'Marks']])
```

Sample Output

	Name	Marks
0	Amit	85
1	Riya	90
2	Rahul	78

22.6 Program to Filter Rows Based on Conditions

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', 'Rahul'],
    'Marks': [85, 90, 78]
}

_df = pd.DataFrame(data)

print(_df[_df['Marks'] > 80])
```

Sample Output

	Name	Marks
0	Amit	85
1	Riya	90

22.7 Program to Find Summary Statistics of a DataFrame

Program

```
import pandas as pd

data = {
    'Marks': [85, 90, 78, 88, 95]
}

_df = pd.DataFrame(data)

print(_df.describe())
```

Sample Output

	Marks
count	5.000000
mean	87.200000
std	6.300794
min	78.000000
25%	85.000000
50%	88.000000
75%	90.000000
max	95.000000

22.8 Program to Add a New Column to a DataFrame

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', 'Rahul'],
    'Marks': [85, 90, 78]
}

_df = pd.DataFrame(data)

_df['Grade'] = ['A', 'A+', 'B']
```

```
print(_df)
```

Sample Output

	Name	Marks	Grade
0	Amit	85	A
1	Riya	90	A+
2	Rahul	78	B

22.9 Program to Handle Missing Values in a DataFrame

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', None],
    'Marks': [85, None, 78]
}

_df = pd.DataFrame(data)

print("Original DataFrame:")
print(_df)

print("After Filling Missing Values:")
print(_df.fillna(0))
```

Sample Output

```
Original DataFrame:
   Name  Marks
0  Amit   85.0
1  Riya   NaN
2  None   78.0
After Filling Missing Values:
   Name  Marks
0  Amit   85.0
1  Riya    0.0
2    0   78.0
```

22.10 Program to Sort Data in a DataFrame

Program

```
import pandas as pd

data = {
    'Name': ['Amit', 'Riya', 'Rahul'],
    'Marks': [85, 90, 78]
}

_df = pd.DataFrame(data)

print(_df.sort_values(by='Marks'))
```

Sample Output

	Name	Marks
2	Rahul	78
0	Amit	85
1	Riya	90

Result

The programs demonstrating the basics of Pandas were successfully executed.

Viva Questions

1. What is Pandas?
 2. What is the difference between Series and DataFrame?
 3. What is the use of read_csv() in Pandas?
 4. How are missing values handled in Pandas?
 5. What is the use of describe() in Pandas?
-

Exercises

1. Create a DataFrame containing employee details.
2. Write a program to filter students scoring more than 75 marks.
3. Write a program to delete a column from a DataFrame.

4. Create a CSV file and read it using Pandas.